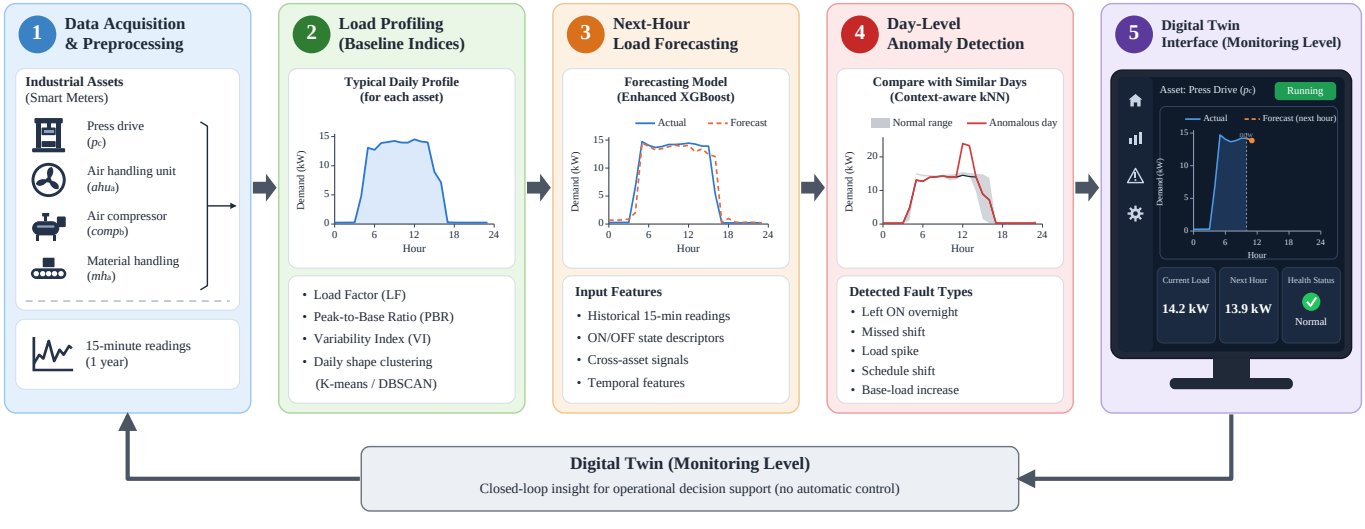

1
Data Acquisition & Preprocessing
Industrial Assets
(Smart Meters)
Press drive (pc_1)
Air handling unit (ahu_8)
Air compressor ($comp_6$)
Material handling (mh_4)
15-minute readings
(1 year)
2
Load Profiling (Baseline Indices)
Typical Daily Profile
(for each asset)
Demand (kW)
Hour
Load Factor (LF)
Peak-to-Base Ratio (PBR)
Variability Index (VI)
Daily shape clustering
(K-means / DBSCAN)
3
Next-Hour Load Forecasting
Forecasting Model
(Enhanced XGBoost)
Actual
Forecast
Input Features
Historical 15-min readings
ON/OFF state descriptors
Cross-asset signals
Temporal features
4
Day-Level Anomaly Detection
Compare with Similar Days
(Context-aware kNN)
Normal range
Anomalous day
Detected Fault Types
Left ON overnight
Missed shift
Load spike
Schedule shift
Base-load increase
5
Digital Twin Interface (Monitoring Level)
Asset: Press Drive (pc_1)
Running
Actual
Forecast (next hour)
now
Current Load
14.2 kW
Next Hour
13.9 kW
Health Status
Normal
Digital Twin (Monitoring Level)
Closed-loop insight for operational decision support (no automatic control)